

```
In [31]: # Library Import
import numpy as np
import pandas as pd
import scipy.stats as stats
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [32]: # Dataset Load
data = pd.read_csv("C:/Users/hanmant/Desktop/sdAssignment/Assignment7/mnist_train.csv")
data.head()
```

```
Out[32]:
```

|   | label | 1x1 | 1x2 | 1x3 | 1x4 | 1x5 | 1x6 | 1x7 | 1x8 | 1x9 | ... | 28x19 | 28x20 | 28x21 | 28x22 |
|---|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------|-------|-------|-------|
| 0 | 5     | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | ... | 0     | 0     | 0     | 0     |
| 1 | 0     | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | ... | 0     | 0     | 0     | 0     |
| 2 | 4     | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | ... | 0     | 0     | 0     | 0     |
| 3 | 1     | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | ... | 0     | 0     | 0     | 0     |
| 4 | 9     | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | ... | 0     | 0     | 0     | 0     |

5 rows × 785 columns



```
In [33]: print("Dataset Shape:", data.shape)
print("\nDataset Info:\n")
data.info()
```

Dataset Shape: (60000, 785)

Dataset Info:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 60000 entries, 0 to 59999
Columns: 785 entries, label to 28x28
dtypes: int64(785)
memory usage: 359.3 MB
```

```
In [34]: data.describe()
```

Out[34]:

|              | label        | 1x1     | 1x2     | 1x3     | 1x4     | 1x5     | 1x6     | 1x7     |
|--------------|--------------|---------|---------|---------|---------|---------|---------|---------|
| <b>count</b> | 60000.000000 | 60000.0 | 60000.0 | 60000.0 | 60000.0 | 60000.0 | 60000.0 | 60000.0 |
| <b>mean</b>  | 4.453933     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     |
| <b>std</b>   | 2.889270     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     |
| <b>min</b>   | 0.000000     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     |
| <b>25%</b>   | 2.000000     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     |
| <b>50%</b>   | 4.000000     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     |
| <b>75%</b>   | 7.000000     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     |
| <b>max</b>   | 9.000000     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     | 0.0     |

8 rows × 785 columns



In [35]:

```
mean_values = data.mean()
std_values = data.std()

print("Mean of Dataset:\n", mean_values.head())
print("\nStandard Deviation:\n", std_values.head())
```

Mean of Dataset:

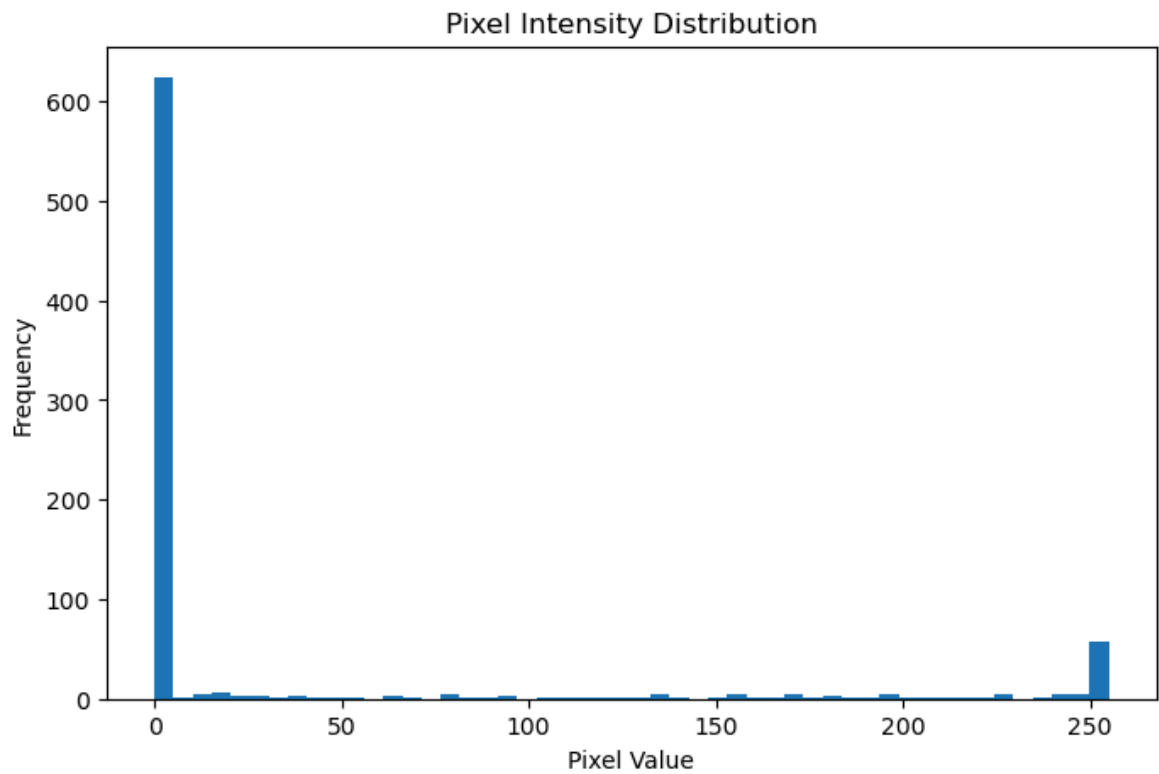
```
label    4.453933
1x1      0.000000
1x2      0.000000
1x3      0.000000
1x4      0.000000
dtype: float64
```

Standard Deviation:

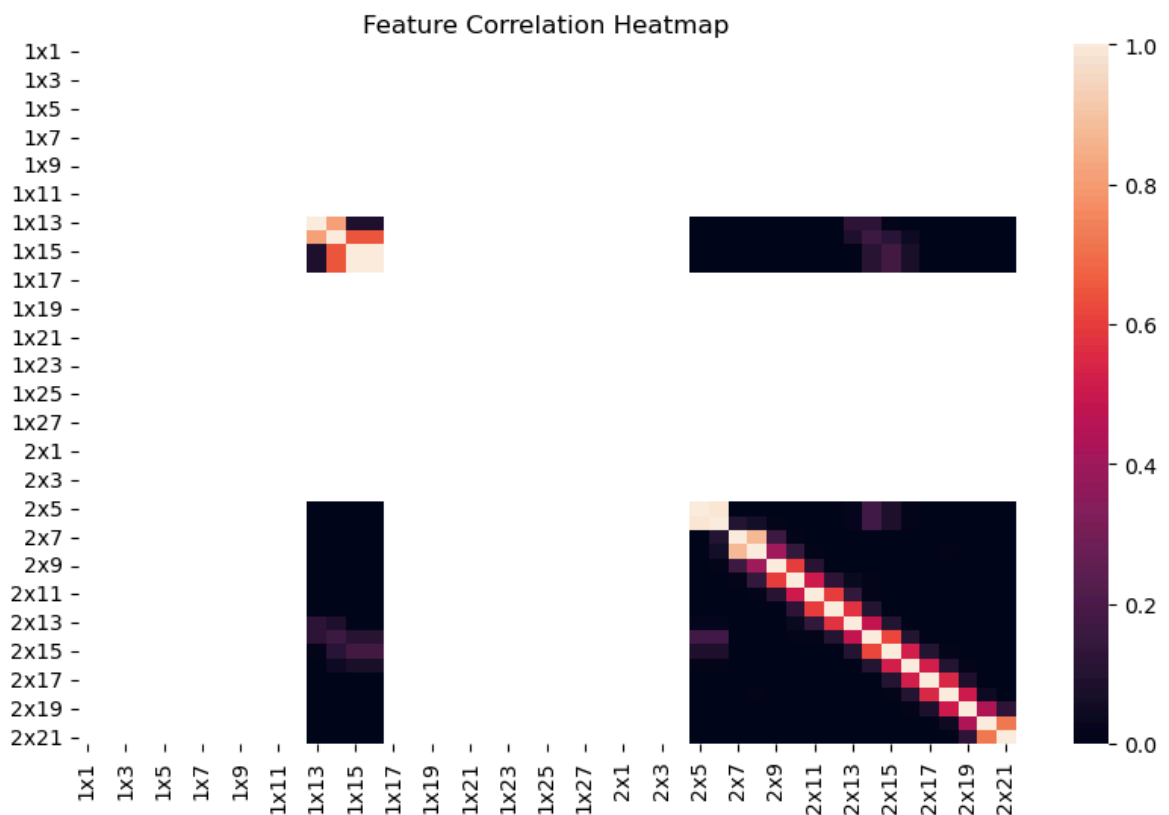
```
label    2.88927
1x1      0.00000
1x2      0.00000
1x3      0.00000
1x4      0.00000
dtype: float64
```

In [36]:

```
plt.figure(figsize=(8,5))
plt.hist(data.iloc[0,1:], bins=50)
plt.title("Pixel Intensity Distribution")
plt.xlabel("Pixel Value")
plt.ylabel("Frequency")
plt.show()
```



```
In [37]: plt.figure(figsize=(10,6))
sns.heatmap(data.iloc[:,1:50].corr())
plt.title("Feature Correlation Heatmap")
plt.show()
```



```
In [38]: plt.figure(figsize=(10,5))
sns.boxplot(data=data.iloc[:,1:20])
plt.title("Pixel Intensity Boxplot")
plt.show()
```

